OFFICIAL BIDDING DOCUMENT

Telecommunications Technology Solutions Inc. (TelTech)

Corporate Tender Secretariat · Procurement & Network Engineering Division

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Author: OLIVER TUNGOL

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Site Survey, RF Planning & Civil/Power Engineering Guidelines

TowerCo Co-Location Standard, Greenfield Siting & Link Budget Engineering Standards

DOCUMENT NUMBER	AUTHOR	TARGET SPECTRUM	BIDDING STATUS
RFP-TELTECH-FWA-2026-001	OLIVER TUNGOL	Band n50 (1.5 GHz, 60 MHz TDD)	Official Request for Proposal

1. ENGINEERING PRINCIPLES & SITING ARCHITECTURE

1.1 Siting Philosophy & Mandatory 3-Tier Sourcing Hierarchy

The primary engineering objective of the 5G FWA rollout is to maximize household population coverage within high-density suburban and rural barangay settlement clusters while minimizing civil capital expenditures and permitting lead times. To achieve rapid 1,000-site Phase 1 execution within 12 months, the Contractor shall execute site delivery strictly following the **Mandatory 3-Tier Infrastructure Sourcing Hierarchy**:

1. **Tier 1 (Mandatory Baseline — Existing TowerCo Co-Location):** Field teams must prioritize co-location on existing operational towers (PTCI/MIDC, EdgePoint, FTAP Globe, FTAP Smart, LDIC, Phil-Tower) located within the designated search ring. The Operator leverages an audited master database of **13,746 operational towers**. Co-location reduces Ready-for-Installation (RFI) lead times from 120–180 days down to 15–30 days while eliminating private land titling and environmental permitting complexities.
2. **Tier 2 (Secondary — TowerCo Build-to-Suit / B2S):** If all existing towers within the search ring exceed structural loading thresholds (EPA limits) or fail RF line-of-sight requirements, the vendor shall engage a licensed Independent TowerCo to execute a Build-to-Suit (B2S) under TelTech's Master Lease Agreement (MLA).
3. **Tier 3 (Restricted Exception — Operator Greenfield Own-Build):** Direct operator-constructed greenfield monopoles are strictly limited to remote isolated barangays located > 2.0 km from any operational tower structure (~9.3% of sites nationwide; 103 sites in Phase 1). Proposing an own-build greenfield site requires formal written submission of **Structural Denial / Space Unavailability Letters** from at least two (2) licensed TowerCos and written sign-off from the TelTech Project Steering Committee.

1.2 Search Ring Specification & Demand Centroid

For each nominal site location assigned in Document 07 (Candidate Site List):

- **Urban & Suburban Search Ring:** Defined as a circle with a radius of **500 meters** centered on the official nominal barangay demand centroid.
- **Rural & Remote Search Ring:** Defined as a circle with a radius of **1,000 meters** (expandable up to 1,500 meters upon TelTech RF approval only if prominent terrain ridges or hills provide superior line-of-sight).
- **Target Coverage Guarantee:** Sited candidate coordinates must achieve unobstructed Line-of-Sight (LOS) and target RSRP coverage (≥ -95 dBm) across ≥ 90% of the inhabited settlement zone.

1.3 Mandatory Three-Candidate Rule (Candidates A, B, C)

For every search ring, the Turnkey Contractor / Site Acquisition Vendor must survey, evaluate, and submit **exactly three (3) distinct candidates** in the Technical Site Survey Report (TSSR):

- **Candidate A (BEST — Primary Recommended):** Optimal TowerCo co-location asset with immediate structural capacity at ≥ 30m AGL, available -48V DC busbar or commercial AC transformer headroom, nearest fiber node, and fastest projected RFI (≤ 21 calendar days).
- **Candidate B (BETTER — Secondary Target / Backup 1):** Fully compliant alternative TowerCo structure within the search ring requiring minor solvable modifications (e.g., dedicated AC sub-meter installation, mount bracket extensions, or secondary fiber lead-in route).
- **Candidate C (GOOD — Contingency / Fallback):** Third viable TowerCo asset or TowerCo Build-to-Suit option, ensuring the search ring is not stalled if Candidates A and B encounter unforeseen structural or commercial deadlocks.
- **14-Day Lease / SAA Escalation Protocol:** If the Site Application & Allocation (SAA) or Service Order (SO) negotiation for Candidate A stalls or exceeds **14 calendar days**, the vendor is contractually mandated to immediately activate and lock **Candidate B** to prevent project schedule slippage.

1.4 Co-Location Candidate Evaluation Criteria Matrix (100-Point Scoring)

- **Space & Structural Capacity (20%):** Antenna mounting space at ≥ 30m to 45m AGL, wind load headroom for 3x n50 antennas + 3x 8T8R RRUs (≤ 45 kg each), and ≥ 2.0m × 2.0m compound ground space for outdoor rectifier/battery cabinets.
- **Access Road & Compound Security (15%):** All-weather paved/concrete road access, established TowerCo security fencing/guarding, and 24/7 unobstructed Operations & Maintenance (O&M) ingress/egress.
- **RF Line-of-Sight & Propagation (25%):** Dominant ground elevation AMSL, clear first Fresnel zone into residential settlement, zero close-in clutter shielding within 200m of sector azimuths (0°, 120°, 240°).
- **Backhaul Transport Proximity (15%):** Proximity to Converge/TelTech fiber distribution closure (Class A ≤ 0.5km, Class B ≤ 1.0km), existing TowerCo cable ladder/conduit entry, or unobstructed northern sky view (0° – 80° elevation arc) for Starlink satellite backhaul.
- **Power Availability & Utility Infrastructure (15%):** Commercial utility grid transformer capacity, available -48V DC busbar tap (≥ 63A) or space for dedicated AC service entrance sub-meter, and shared backup genset/battery support.
- **Lease & Commercial Terms (10%):** Fast-track Service Order execution under existing TowerCo Master Lease Agreement (MLA), monthly recurring fee within budget, and guaranteed RFI timeline ≤ 30 calendar days.

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2. DETAILED TECHNICAL SITE SURVEY REPORT (TSSR) REQUIREMENTS

For every nominal site candidate, the Contractor shall complete and submit a comprehensive TSSR conforming to the following structure:

2.1 General Site Identification & Geography

- Site Name, Candidate ID, Official PSGC Code, Region, Province, Municipality, and Barangay.
- Precision WGS84 GPS Coordinates (Latitude, Longitude recorded with >= 6 decimal places via dual-frequency GPS receiver).
- Ground Elevation Above Mean Sea Level (AMSL in meters).
- Access Road Assessment: Road surface type (concrete, asphalt, macadam, dirt trail), bridge weight limits, and 4x4 hauling accessibility during monsoon seasons.

2.2 Tower Structure & Loading Audit (Co-Location Sites)

- Operating TowerCo Entity (PTCI/MIDC, FTAP Globe, EdgePoint, FTAP Smart, LDIC, FTAP BTS).
- Tower Type (Self-Supporting 3-Leg/4-Leg Tower, Monopole, Guyed Mast, Rooftop Tower).
- Tower Height Above Ground Level (AGL in meters).

- Structural Space Audit: Proposed mounting height for 3x Band n50 antennas (recommended centerline: 30m to 45m AGL).
- Structural Loading Verification: Confirmation that existing tower members and foundation can accommodate additional 3x 8T8R RRUs/AAUs (< 45 kg each) and wind load profile of 250 km/h.
- Panoramic Drone Photographic Survey: High-resolution 360° continuous panoramic video and 12-cardinal-point still photos at proposed antenna centerline height to verify optical line-of-sight across populated settlements.

2.3 Power Infrastructure & Utility Audit

- Existing Commercial AC Utility Power: Electric Cooperative / Distribution Utility name (e.g., MERALCO, PELCO, BENECO, CAPELCO, ILPI).
- Service Entrance Capacity: Transformer rating (kVA), service voltage (230V / 400V, 60 Hz), breaker rating, and phase balance.
- TowerCo DC Power Availability: Sourcing from TowerCo DC busbar (-48V DC) vs. installing Operator-dedicated rectifier subrack.
- Sub-metering feasibility and cable tray routing for power cables.
- Grounding System Audit: Measurement of existing earth ground resistance using a 3-point fall-of-potential earth tester. Earth resistance must be <= 5.0 Ohms.

2.4 Backhaul Transmission Path Survey

- Audited Converge Fiber Network Match (subject to verification on the ground): Distance to nearest confirmed node candidate (Class A <= 0.5 km, Class B <= 1.0 km) or line-tap closure (Class C <= 0.5 km).
- Cable Entry Facility: Manhole, handhole, or aerial pole lead-in routing into tower shelter/compound.
- For Remote Satellite Sited Candidates (Starlink Business): Satellite antenna mounting location (unobstructed sky view with 0° to 80° northern elevation arc for Starlink constellation tracking).

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3. RF PROPAGATION & CELL DIMENSIONING GUIDELINES

3.1 Dual-Tier Spectrum Architecture & Operating Bandwidth (Band n50)

In strict alignment with **Bid Bulletin No. 1 (Section 2.1)** and NTC frequency authorizations, the 5G FWA network operates on a dual-tier spectrum deployment model governed by regional utility franchise boundaries:

1. **Metro Manila (NCR / Meralco Franchise Footprint):**
   - **Operating Frequency Range: 1457.0 – 1517.0 MHz (60.0 MHz continuous TDD carrier)**.
   - **Center Frequency: 1487.0 MHz**.
   - **Transmission Bandwidth: 162 Resource Blocks (RBs)** at 30 kHz Subcarrier Spacing (SCS).
   - *Spectrum Boundary Condition:* The 1437.0 – 1457.0 MHz sub-band is assigned to Meralco within its power distribution franchise area and is strictly excluded from Operator transmission in NCR. Digital filtering within the RRU transceiver enforces sharp attenuation at 1457.0 MHz.
2. **Rest of the Country (Provincial / Regional / Non-Meralco Areas):**
   - **Operating Frequency Range: 1437.0 – 1517.0 MHz (80.0 MHz continuous TDD carrier — Desired Baseline)**.
   - **Center Frequency: 1477.0 MHz**.
   - **Transmission Bandwidth: 217 Resource Blocks (RBs)** at 30 kHz SCS.
   - *Capacity & Throughput Advantage:* In all provincial municipalities and rural barangays outside the Meralco territory, the full 80 MHz channel is activated, unlocking +34% additional physical resource blocks and driving aggregate sector downlink capacity to 650 – 720 Mbps per site.
3. **Mandatory Hardware Tuning & Filtering Specifications:**

- **Native Hardware IBW:** All proposed 5G NR Baseband Units (BBU) and Remote Radio Units (RRU) / Active Antenna Units (AAU) must natively support the full 3GPP Band n50 tuning range (**1432.0 – 1517.0 MHz**) in hardware.
- **Software-Defined Bandwidth:** Transceivers must support dynamic software reconfiguration across 40 MHz, 50 MHz, 60 MHz, and 80 MHz without physical component replacement or tower climbs.
- **Adjacent Satellite Protection:** Transceivers must comply with **ITU-R M.2159-0 / WRC-15 Resolution 223** to protect adjacent Maritime Mobile Satellite Service (MSS, 1518.0 – 1525.0 MHz). Unwanted out-of-band emissions at 1517.0 MHz must not exceed -30 dBm/MHz at the antenna port (3GPP TS 38.104 Category B).

3.2 Link Budget Parameters (Band n50, 1.5 GHz)

PARAMETER	UNIT	60 MHZ MODE (NCR / MERALCO)	80 MHZ MODE (PROVINCIAL / RURAL)	TECHNICAL NOTES & MODELING BASIS
Operating Frequency	MHz	1487.0 (Center)	1477.0 (Center)	Band n50 TDD
Channel Bandwidth	MHz	60.0 MHz	**80.0 MHz**	162 RBs (60 MHz) vs. **217 RBs (80 MHz)**
Subcarrier Spacing (SCS)	kHz	30.0	30.0	5G NR Numerology μ=1
BTS Maximum Conducted Power	dBm / W	55.0 dBm (320 W)	55.0 dBm (320 W)	8 x 40W RRU configuration
BTS Power per Resource Block	dBm	32.9 dBm	**31.6 dBm**	Uniform power allocation across active RBs
BTS Antenna Gain	dBi	17.5 dBi	17.5 dBi	8-port cross-pol macro antenna
BTS Cable / Jumper Loss	dB	1.0 dB	1.0 dB	Low-loss jumper interface
CPE Maximum Transmit Power	dBm / W	23.0 dBm (200 mW)	23.0 dBm (200 mW)	Power Class 3 UE (or Class 2 26 dBm)
CPE Antenna Gain	dBi	9.0 dBi	9.0 dBi	Baseline model (6 dBi indoor / 12 dBi outdoor)
CPE Body / Indoor Penetration Loss	dB	8.0 dB	8.0 dB	Concrete hollow block & glass attenuation
Foliage / Clutter Margin	dB	6.0 dB	6.0 dB	Typical rural tropical tree canopy
Shadowing Margin (Log-Normal)	dB	7.0 dB	7.0 dB	90% edge cell coverage reliability
Thermal Noise Density	dBm/Hz	-174.0	-174.0	kT @ 290K
Receiver Noise Figure	dB	6.0 dB (CPE) / 3.5 dB (gNB)	6.0 dB (CPE) / 3.5 dB (gNB)	Active RF front-end
Target SINR for Min Service	dB	-1.0 dB (QPSK 1/3)	-1.0 dB (QPSK 1/3)	Cell-edge SLA threshold (+10 dB for 64QAM)
Maximum Allowable Path Loss (MAPL)	**dB**	**139.8 dB**	**138.5 dB**	**Uplink coverage limiting (Δ ≈ 1.3 dB for 80 MHz)**
Calculated Reliable Cell Radius	**km**	**3.2 km (Suburban)**	**3.0 km (Rural)**	**Planning assumption in Philippine clutter**
Peak Aggregate Sector DL Throughput	**Mbps**	**~480 – 520 Mbps**	**~650 – 720 Mbps**	**Multi-user MIMO, 256QAM downlink**

3.3 Nominal Sector Azimuth & Tilt Strategy

- **Sector Count:** 3 sectors per macro site with 120° nominal separation (0°, 120°, 240°). Azimuths must be tailored during TSSR to track actual barangay population concentrations along transport corridors and coastal plains.
- **Electrical Remote Downtilt (eRET):** Antennas must feature AISG 2.0 remote tilt adjustable between 2° and 12° from the central OSS. Mechanical tilt shall only be applied when required to clear close-in obstructions (>15°).

4. CIVIL WORKS & STRUCTURAL ENGINEERING (CLASS T4 GREENFIELD & TOWERCO BTS SITES)

For sites requiring self-constructed greenfield towers or TowerCo Build-to-Suit (B2S) structures, the Contractor shall adhere to the following civil and structural standards conforming to the **National Structural Code of the Philippines (NSCP 2015 7th Edition, Section 207 Wind Loads)** and **TIA-222-G/H**:

4.1 Philippine Basic Wind Speed Regimes & Wind Zone Mapping

Structural calculations and wind force determinations must strictly comply with NSCP 2015 basic wind speed maps:

- **Wind Zone 1 (Ultimate Wind Speed V = 315 km/h / 87.5 m/s):** Covers the high-typhoon corridor along the eastern seaboard (Batanes, Cagayan, Isabela, Aurora, Quezon/Polillo, Bicol Region, Samar, Leyte, Dinagat Islands, Surigao del Norte/Sur, Davao Oriental).
- **Wind Zone 2 (Ultimate Wind Speed V = 270 km/h / 75.0 m/s):** Covers Central and Western Luzon, National Capital Region (NCR), Central Luzon (Pampanga, Bulacan, Tarlac, Bataan, Zambales, Nueva Ecija), CALABARZON, MIMAROPA, and Western/Central Visayas (Iloilo, Cebu, Bohol, Negros).
- **Wind Zone 3 (Ultimate Wind Speed V = 200 km/h / 55.6 m/s):** Covers low-risk regions including Palawan, Zamboanga Peninsula, and southern inland Mindanao.

4.2 Structural Limit States & Deflection Criteria

- **Survivable Wind Speed ($V_{survival}$ — Ultimate Limit State / LRFD):** Designed for **270 km/h to 315 km/h** 3-second gust wind speed under Ultimate Limit State with full equipment loading (3x n50 antennas, 3x RRUs, satellite dish, and mounting steelwork). Zero plastic collapse or structural overturning.
- **Operational Wind Speed ($V_{operational}$ — Serviceability Limit State / SLS):** Designed for **140 km/h to 160 km/h (40.0 m/s to 44.4 m/s)** continuous operational wind speed (1.0 load factor per TIA-222). Structure maintains rigid RF alignment, ensuring uninterrupted 5G beamforming, MIMO rank-4 transmission, and backhaul link availability (≥ 99.995%).
- **Allowable Twist (θ_{twist} / Torsional Rotation):** ≤ ±0.5° (30 arcmin) at 30m AGL for narrow-beam / microwave dish antennas; ≤ ±1.0° (60 arcmin) at 30m AGL for 5G Band n50 macro sectors.
- **Allowable Sway / Tilt (θ_{sway} / Angular Deflection):** ≤ ±1.0° (60 arcmin) at top antenna mounting centerline (30.0m AGL).
- **Allowable Verticality (Tower Plumbness / Alignment):** ≤ 1 : 500 (0.20% of Tower Height H) throughout the full mast height under dead load and still air. For a 30.0m monopole, maximum horizontal deviation $\Delta \leq 60$ mm (for 40.0m monopole, $\Delta \leq 80$ mm). Tolerance between consecutive slip-joints: ≤ 1 : 1000 (0.10%).
- **Horizontal Service Deflection (Δ_{top}):** ≤ H / 100 (300 mm maximum at 30.0m AGL).

4.3 Monopole Tower & Material Specification

- **Tower Structure:** 30-meter or 40-meter polygonal tubular steel monopole constructed in 3 to 4 slip-joint sections with minimum 1.5 × inside diameter overlap.
- **Structural Steel:** ASTM A572 Grade 50 (F_y = 345 MPa / 50 ksi, F_u = 450 MPa).
- **Corrosion Protection:** Hot-dip galvanized per ASTM A123 / ISO 1461 with minimum zinc coating thickness ≥ 85 µm (610 g/m²).
- **Fasteners & Anchors:** Anchor bolts ASTM F1554 Grade 55/105 with double heavy-hex nuts; structural bolts ASTM A325 / ISO 898-1 Class 8.8 hot-dip galvanized.

4.4 Compound Civil Works & Foundation

- **Compound Footprint:** Standard 8.0m × 8.0m (64 m²) gravel-topped compound with geotextile weed barrier.

- **Perimeter Fencing:** 2.4-meter high cyclone wire mesh fence topped with 3 strands of galvanized concertina razor wire.
- **Equipment Pad:** Reinforced concrete equipment plinth (Class A concrete, 28-day compressive strength $f'_c \geq 28$ MPa / 4,000 psi, ASTM A615 Grade 60 rebar) for outdoor power/battery cabinets.
- **Foundation:** Reinforced concrete pad and chimney mat foundation or bored piles engineered with Safety Factor ≥ 2.0 against soil bearing failure and ≥ 1.5 against overturning/sliding.

4.5 Lightning Protection & Earthing Grid

- Top-mounted Franklin lightning air terminal extending ≥ 2.0m above uppermost antenna.
- Down-conductor: 70mm² bare stranded copper cable routed externally and clamped every 1.0m to tower shaft.
- Ground ring: Continuous buried bare copper loop around tower base and compound with copper-clad steel ground rods (19mm dia × 3.0m length) exothermically welded (Cadweld) at all junctions, achieving earth ground resistance ≤ 5.0 Ohms.

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5. SINGLE SITE VERIFICATION (SSV) & CLUSTER ACCEPTANCE CRITERIA

Before any site is granted Commercial On-Air Acceptance, the Contractor must complete and submit a signed SSV Report verifying:

1. **Stationary Functional Tests:**

* Sector Verification: Physical azimuth and tilt match TSSR approval within +/- 2°.
* RF Transmit Power: Conducted power on all 8 transmit paths within +/- 0.5 dB of nominal.
* VSWR / Return Loss: Return loss >= 15 dB across all antenna feeder lines.
* Max Throughput Test: Single CPE in line-of-sight achieving >= 300 Mbps DL and >= 50 Mbps UL (aggregate sector peak >= 450 Mbps DL / >= 60 Mbps UL verified under multi-CPE or loaded test).

2. **Short Drive Test & Static Multi-Point Test:**

* 5 stationary measurement points per sector (Near: 500m, Mid: 1.5 km, Far/Edge: 3.0 km).
* RSRP >= -95 dBm across >= 90% of surveyed barangay inhabited zone.
* Ping RTT <= 18 ms target (<= 30 ms maximum contractual limit) to local UPF IP address over fiber backhaul; packet loss <= 0.10% over 1,000 continuous pings.
* Inter-sector handover execution without drop during transition.

(End of Appendix D — Site Survey, RF Planning & Civil Engineering Guidelines)
